6COSC020W Applied AI

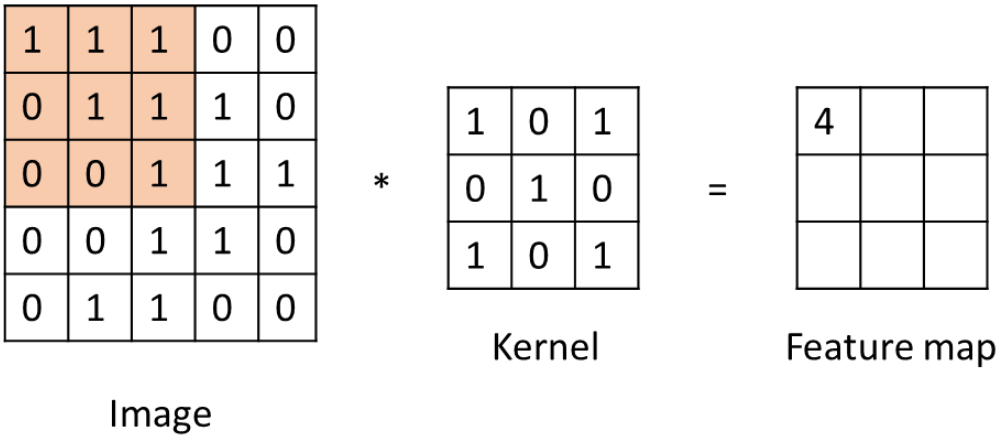
Tutorial Week 10: Deep Learning and CNNs

Aim**:**

- Review key deep learning concepts.
- Calculate the results of a convolution.
- Calculate the result of max pooling and average pooling.
- Train and evaluate a CNN network
- Understand the performance of the network
- Improve the performance of the network

1.- Define in your own words the following terms: deep learning, classification, features, filters, convolution, CNN, convolution layer, pooling layer, ReLu, computer vision.

2.- Calculate the result of the following convolution. The kernel is 3x3, stride is 1.



3.- Given the following feature map, **calculate:**

29	15	28	184
0	100	70	38
12	12	7	2
12	12	45	6

a) The result after applying a 2x2 **max pooling** with a stride of 2 pixels.

b) The result after applying a 2x2 **average pooling** with a stride of 2 pixels.

4.- Practice: Tutorial_Week10.ipynb. CNNs.

1) Download the file *Tutorial_Week10.ipynb* from Blackboard
2) Upload and open the file with Google Collaboratory (recommended) or Anaconda (selecting Jupyter Notebook – you may need to install TensorFlow).
3) Work through the Jupyter Notebook. Answer the questions and do the exercises.